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When it comes to security, all of us are paranoid, and with good reason. With millions of ways to exploit our computers or data, and hundreds of thousands of people trying to do so, can you blame us? But how many of us are *really* careful, or even know what to do when the bits hit the fan? This workshop will help you do just that...

Am I Infected?

So how do you know if your computer is infected with a virus? Here are some indications:

- It runs considerably slower than usual
- It frequently stops responding
- It crashes and restarts
- Applications don't properly
- One or more disks are inaccessible
- You see distorted windows and dialog boxes
- Files get deleted for no reason
- You get strange error messages
- You see strange behaviour in your e-mail program

What Are The Kinds Of Virus Scanners?

Anti-virus software are of two types—those that can be invoked when needed, and those that are always active in the background.

On-Access Scanners

Also known as Memory Resident Scanners, these run in the background when the PC is running. The function of an on-access scanner is to monitor all activity on your machine such as files being read or accessed, downloads, receiving, sending or reading e-mail, and so on.

If, while monitoring activity, the on-access scanner encounters a file whose activity it identifies as virus-like (such as a file trying to dial a certain number or trying to e-mail something from your PC), it informs you or takes some kind of action (depending on how you've configured it).

Since they always run in the background and constantly access files, on-access scanners invariably slow down your system, at least to some extent and depending on your system specs. Most on-access scanners allow you to disable on-access scanning temporarily. If, for example, you need to run a resource-hungry application such as a 3D game, you might want to disable on-access scanning, provided you're not connected to the Internet and are not accessing files from removable media.

You should never install two on-access scanners on your machine because each of them is likely to detect the other as a virus, and this can even cause your system to crash.

On-Demand Scanners

These scan files or folders at the user's request. All anti-viruses are on-demand scanners; some are both on-access as well as on-demand. On-demand scanners are useful when you want to scan a file or folder where you want to make sure there isn't a virus. For example, if you're burning a CD and giving it to a friend, it's a good idea to scan it first.

You may install as many on-demand scanners as you wish. Having more than one can be a good thing—two anti-viruses can detect more viruses! You should, however, remember to run on-demand scanners separately. Also, take care before you run an on-demand scanner that an on-access scanner is not already running.

What Is Heuristic Scanning?

All anti-virus software depend on virus definition files which contain virus "signatures" to identify a virus. This is traditional scanning and here, the anti-virus looks through files and searches for strings that resemble the signature of a known virus. But newer viruses are always being unleashed, and their signatures obviously won't exist in the database. What does the anti-virus do in that case? It falls back on its heuristic scanning capabilities.

The heuristic scan capability of an anti-virus is its ability to detect an undefined virus from its suspicious activities. It works on the principle that viruses usually use certain methods of infecting other files, and if an application is found to be using such methods—or exhibiting such typical behaviour—it is detected as a virus.

A file can also be wrongly detected to be a virus. This happens more often when the heuristic "sensitivity" is kept high. You need to decide the course of action in such a case by taking a look at the filename. There's a trade-off here: if you keep the sensitivity high, you're safer, but you'll get more false alarms.

What Is A firewall?

All Internet communication takes place by exchange of individual packets of data. Each packet is transmitted by a source machine towards a destination machine. When the latter receives the packet, it sends an acknowledgement packet to the source machine that the packet has reached it. For the receiving computer to know who sent it the acknowledgement packet, it must also contain the IP address and port number of the source machine.